

Justin Doty, Ph.D.

☎ +1 (714) 833-0867 ✉ justindoty@gmail.com 🌐 justdoty.github.io 📄 github.com/justdoty

PROFILE

Senior Application Engineer at MathWorks specializing in econometric modeling. Ph.D. economist with research expertise in econometrics, empirical industrial organization, and differential privacy, including over three years developing formally private statistical methods at the U.S. Census Bureau.

EXPERTISE

Theoretical and Applied Econometrics · Empirical Industrial Organization · Heterogeneous Treatment Effects · Measurement Error · Quantile Regression · Productivity Estimation · Auctions · Differential Privacy

EDUCATION

The University of Iowa, Iowa City, IA 2016–2022
Ph.D. in Economics
Dissertation: *Identification and Estimation of Production Functions with Unobserved Heterogeneity*
Advisor: *Suyong Song*
M.A. in Economics
California State University, Fullerton (CSUF), Fullerton, CA 2011–2016
B.A. in Economics, minor in Mathematics

EXPERIENCE

MathWorks, Chevy Chase, MD Feb 2026–present
Senior Application Engineer – Econometric Modeling, Finance Division

- Deliver technical demonstrations and consultations to economists on econometric modeling in MATLAB

U.S. Census Bureau, Washington, DC Aug 2022–Jan 2026
Research Economist, Disclosure Avoidance Research Group

- Modernized statistical disclosure methodology for protecting data confidentiality
- Developed formally private algorithms for heavy-tailed, establishment-level business data
- Built and tested statistical programs that protect categorical data using local differential privacy and correct inferences made on privacy-protected data
- Tested experimental iterations of the Disclosure Avoidance System (DAS)
- Co-authored research developing formally private algorithms for a range of data release scenarios
- Reviewed and approved internal working papers for public release
- Provided subject-matter expertise on statistical disclosure methods for planned external data releases

The University of Iowa, Iowa City, IA 2016–2022
Graduate Teaching Assistant

- Taught discussion sections and graded for courses including *Ph.D. Econometrics*, *Ph.D. Applied Econometrics*, *Ph.D. Statistics for Economics*, *Economic and Business Forecasting*, *Statistics for Strategy*, *Managerial Economics*, and principles of micro- and macroeconomics
- Provided supplemental instruction in econometric theory and statistical programming to graduate and undergraduate students

PUBLISHED PAPERS

Finley, Caruso, Doty, Machanavajjhala, Meyer, Pujol, Sexton, and Turner. (2026). “Slowly Scaling Per-Record Differential Privacy.” *Journal of Privacy and Confidentiality* 16 (1). <https://doi.org/10.29012/jpc.992>

WORKING PAPERS

“Technology-Type-Heterogeneous Elasticities in Gross-Output Production Functions” (*with Suyong Song*)
“A Dynamic Framework for Identification and Estimation of Nonseparable Production Functions”
“Estimating Quantile Production Functions: A Control Function Approach” (*with Suyong Song*)

WORKS IN PROGRESS

“Multinomial Randomized Response for Data Sharing” (*with Caleb Floyd and Rolando Rodriguez*)
“Identification of Gross-Output Production Functions with Mismeasured Inputs”
“Estimating Industry Competitor Network Effects on Firm-Level Productivity”

AWARDS AND FELLOWSHIPS

Ballard and Seashore Dissertation Fellowship, University of Iowa	2021
Summer Fellowship, University of Iowa	2020
Post-Comprehensive Research Fellowship, University of Iowa	2020
GSS/Graduate College Presentation Travel Funds, University of Iowa	2019
Stewart-Long Graduate Study Award, CSUF	2015
Outstanding Student Scholarly and Creative Activities Award, CSUF	2015

CONFERENCE PRESENTATIONS

American Economic Association (AEA/ASSA) Annual Meeting	January 2025
North America Summer Meeting of the Econometric Society (NASMES)	June 2022
European Winter Meeting of the Econometric Society (EWMES)	December 2021
Southern Economic Association (SEA)	November 2021
Joint Statistical Meetings (JSM)	August 2020
Midwest Econometrics Group (MEG)	October 2019

SKILLS

Programming & Scripting: MATLAB (Econometrics, Statistics and Machine Learning, Optimization Toolboxes), R (tidyverse, data.table, quantreg, parallel, rstan), Python (pandas, numpy, matplotlib, scikit-learn, pytorch, pyspark)
Tools & Environments: Git, GitHub, Jupyter Notebooks, AWS EC2, Gurobi, Stan, Vim
Concepts: Linear, Logistic, and Multinomial Regression, Nonlinear Models, Kernel Density Estimators, Monte Carlo Simulations, Large-Sample Theory, Machine Learning Algorithms, Time Series Forecasting, Small Area Estimation, Differential Privacy and Formal Privacy Variants, Database Reconstruction and Membership Inference Attacks

CERTIFICATIONS AND TRAININGS

Machine Learning Specialist – U.S. Census Data Science Training Program	2025
Global Leadership – Student Leadership Institute, CSUF	2016

JOURNAL REFEREE EXPERIENCE

Empirical Economics